E COMMERCE SALES ANALYSIS & PERFORMANCE DASHBOARD USING EXCEL

Project Objective:

- The main objective of this project is to analyze sales data to gain insights into customer purchasing patterns
- Top selling products, and store performance.
- The project involves Data Cleaning, Summary Statistics.
- Data using to VLOOKUP/XLOOKUP.
- Creating Interactive Dashboard with Pivot Tables and Pivot Charts to support business decision making.

Dataset Description:

The dataset, sales data recorded across various retail stores between 2023 and 2025. That information about each transaction, including the order date, customer ID, product ID, store location, purchased quantity, unit price, applied discount, and total calculated amount. Additionally relational lookup operations (XLOOKUP) used to merge customer names and product names from secondary dimension tables to improve the dataset for analytical purposes.

Column Description

Column Name	Description
Order Date	The date on which the customer placed the order
Customer ID	Unique identifier assigned to each customer
Product ID	Unique identifier assigned to each product
Store ID	Unique identify assigned to each store
Quantity	Number of units purchased in the transaction
Unit price	The price per single unit of the product
Discount	Discount rate applied to the order
Total Amount	Final sales amount calculated after applying quantity, price, and discount
Customer Name	Full name of the customer, using XLOOKUP based on Customer ID
Product Name	Product Name of the item purchased, using XLOOKUP.
Store Name	Full Name of the store, using XLOOKUP based on Store ID

Data cleaning & transformation:

Formula used:

- REMOVE DUPLICATE DATA: Data>Data Tool>Remove Duplicates

- CORRECT FORMATTING: (CUSTOMER DIM) PROPER(TRIM(CLEAN(B2)
- PROPER(TRIM(CLEAN(E2) PROPER(TRIM(CLEAN(F2)
  PROPER(TRIM(CLEAN(H2)

- CORRECT FORMATTING: (PRODUCT DIM) PROPER(TRIM(CLEAN(B2)
  PROPER(TRIM(CLEAN(C2) PROPER(TRIM(CLEAN(D2)
  PROPER(TRIM(CLEAN(E2)

- CORRECT FORMATTING: (STORE DIM) PROPER(TRIM(CLEAN(B2)
  PROPER(TRIM(CLEAN(D2)

- LOYALTY_LEVEL COLUMN HANDLE MISSING DATA:
  IF(ISBLANK(H2),"UNKNOWN",H2)

- COST AND STOCK COLUMN HANDLE MISSING DATA:
  COST: IF(ISBLANK(F2),AVERAGE(F2:F103),F2)
  STOCK:IF(ISBLANK(G2),AVERAGE(G2:G103),G2)

- QUANTITY, UNIT PRICE AND DISCOUNT COLUMNS HANDLE MISSING DATA with AVERAGE:
  Quantity: IF(ISBLANK(F2),AVERAGE(F2:F2001),F2)
  Unit price: IF(ISBLANK(G2),AVERAGE(G2:G2001),G2)
  DISCOUNT:IF(ISBLANK(H2),AVERAGE(H2:H2001),H2)

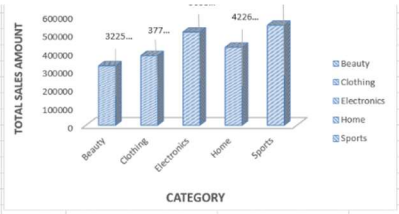
With screenshot:

DESCRIPTIVE STATISTICS

DESCRIPTIVE STATISTICS			
Quantity		Unit_Price	
Mean	2.4765	Mean	513.9831146
Standard Error	0.024890433	Standard Error	6.307795632
Median	2	Median	522.43
Mode	2	Mode	871.34
Standard Deviatior	1.113133992	Standard Deviatic	282.0931964
Sample Variance	1.239067284	Sample Variance	79576.57147
Kurtosis	-1.347725858	Kurtosis	-1.189707573
Skewness	0.029733624	Skewness	-0.036751141
Range	3	Range	979.8
Minimum	1	Minimum	20.09
Maximum	4	Maximum	999.89
Sum	4953	Sum	1027966.229
Count	2000	Count	2000

1. Quantity:
Mean (Average): 2.48
Median & Mode: 2
Range: 1 to 4 (Range: 3)
Total Sum: 4,953
2. Unit Price:
Mean (Average): 513.98
Median: 522.43
Mode: 871.34
Range: 20.09 to 999.89 (Range: 979.8)
Total Sum: 1,027,966.23
Overall Takeaway:
Across 2,000 recorded transactions, the average purchase quantity is 2.48 units (ranging from 1 to 4 units) at an average unit price of 513.98, A total value of 1,027,966.23.

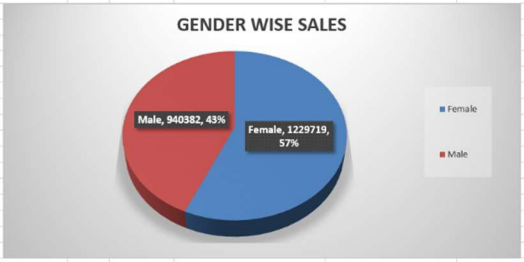
Beauty	322382
Clothing	377543
Electronics	505315
Home	422651
Sports	542010
Grand Total	**2170101**



Row Labels	Average of Total_Amount
Aguirre-Keller	93048
Alexander Group	118814
Bean, Jones and	158904
Benson-Elliott	115363
Benson-Elliott	115363
Black-Horn	98976
Frederick-Wright	111705
Gardner, Humphr	114593
Gray Ltd	80510
Guerrero-Schmitt	105437
Hill, Wagner and	109792
Holloway Ltd	120934
Johnson-Wright	123860
Mann-Austin	94226
Martinez, Adkins	99629
Martinez-Mullins	107987
Mcdonald Inc	106248
Moore, Padilla a	108429
Rivera-Warner	85139
Shelton PLC	99679
Thomas, Bridges	116829
Grand Total	**2170101**



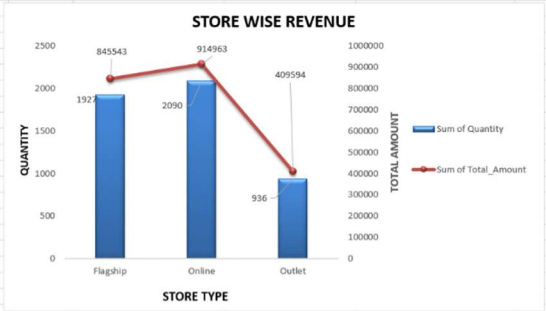
Row Labels	Sum of Total_Amount
Female	1229719
Male	940382
Grand Total	**2170101**



Sum of Total_Amount	Column Labels			
	2023	2024	2025	Grand Total
Row Labels				
East	68723	291457	264367	624548
North	72797	337185	231459	641440
South	62756	332830	237918	633503
West	18105	130810	121694	270609
Grand Total	**222381**	**1092282**	**855438**	**2170101**



Row Labels	Sum of Quantity	Sum of Total_Amount
Flagship	1927	845543
Online	2090	914963
Outlet	936	409594
Grand Total	**4953**	**2170101**

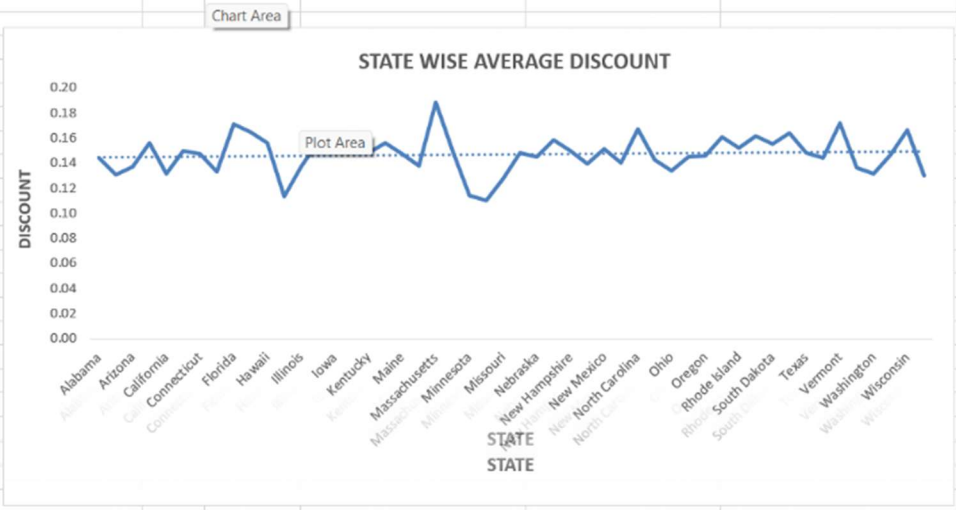


Row Labels	Sum of Quantity
Bronze	1209
Gold	1189
Platinum	1485
Silver	1034
UNKNOWN	36
Grand Total	**4953**



Row Labels	Average of Discount

Row Labels	Average of Discount
Alabama	0.14
Alaska	0.13
Arizona	0.14
Arkansas	0.16
California	0.13
Colorado	0.15
Connecticut	0.15
Delaware	0.13
Florida	0.17
Georgia	0.17
Hawaii	0.16
Idaho	0.11
Illinois	0.14
Indiana	0.16
Iowa	0.16
Kansas	0.16
Kentucky	0.15
Louisiana	0.16
Louisiana	0.16
Maine	0.15
Maryland	0.14
Massachusetts	0.19
Michigan	0.15
Minnesota	0.11
Mississippi	0.11
Missouri	0.13
Montana	0.15
Nebraska	0.15
Nevada	0.16
New Hampshire	0.15
New Jersey	0.14
New Mexico	0.15
New York	0.14
North Carolina	0.17
North Dakota	0.14
Ohio	0.13
Oklahoma	0.14
Oregon	0.15
Pennsylvania	0.16
Rhode Island	0.15
South Carolina	0.16
South Dakota	0.16
Tennessee	0.16
Texas	0.15
Utah	0.14
Vermont	0.17
Virginia	0.14
Washington	0.13
West Virginia	0.15
Wisconsin	0.17
Wyoming	0.13
Grand Total	**0.15**



Xlookup Using and Creating Table

Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	TOTAL AMOUNT	Customer Name	Product_Name	Store Name
07-07-24	CUST1409	PROD97	STORE3	1	218	0.09	198.42	Sarah Smith	Cricket Bat	Johnson-Wright
25-12-23	CUST1467	PROD74	STORE17	3	104	0.27	228.79	Michael Foley	Bluetooth Car Audio Syster	Holloway Ltd
03-06-24	CUST1056	PROD49	STORE8	4	914	0.25	2742.15	Christopher Sutton	Men's Roll-On Perfume	Bean, Jones and Park
29-11-23	CUST1346	PROD76	STORE8	1	922	0.16	774.07	John Jordan	Wireless Mouse	Bean, Jones and Park
06-01-24	CUST1493	PROD59	STORE5	3	745	0.18	1833.44	Amanda Powell	Men's Hair Styling Gel	Gray Ltd
26-07-24	CUST1116	PROD15	STORE7	3	147	0.19	357.23	Anthony Phillips	4K Home Projector	Martinez-Mullins
12-07-25	CUST1016	PROD56	STORE4	2	785	0.15	1335.03	John Scott	Premium Blanket Throw	Alexander Group
18-11-23	CUST1152	PROD70	STORE8	4	342	0.23	1054.56	Patrick Roberts	Kitchen Storage Rack	Bean, Jones and Park
04-05-25	CUST1187	PROD99	STORE5	1	448	0.07	416.87	Dr. Suzanne Bullock	Stainless Steel Cookware Se	Gray Ltd
31-07-25	CUST1464	PROD33	STORE0	2	873	0.23	1344.68	David Pierce	Modern Wall Clock	Rivera-Warner
13-02-24	CUST1065	PROD24	STORE14	1	279	0.07	259.13	Michael Thomas	Foldable Camping Chair	Mcdonald Inc
24-05-25	CUST1320	PROD39	STORE1	4	287	0.17	952.57	Jessica Gonzalez	Portable Power Bank	Martinez, Adkins and Johnson
24-08-24	CUST1237	PROD88	STORE4	4	645	0.01	2552.54	Wesley Short	Sports Wrist Sweatbands	Alexander Group
06-08-25	CUST1443	PROD34	STORE16	2	317	0.30	443.66	Adam Castro	Gaming Mechanical Keyboa	Frederick-Wright
21-06-25	CUST1348	PROD24	STORE15	4	403	0.29	1145.85	Sherry Cabrera	Foldable Camping Chair	Shelton PLC
10-12-24	CUST1435	PROD36	STORE9	1	871	0.04	836.49	Lindsay Beasley	Women's Summer Dress	Moore, Padilla and Butler
13-09-25	CUST1276	PROD65	STORE19	2	452	0.25	677.97	Arthur Warren	Adjustable Dumbbells 20kg	Mann-Austin
11-12-24	CUST1079	PROD27	STORE4	2	503	0.23	774.14	Molly Gardner	Decorative Floor Vase	Alexander Group
13-08-24	CUST1094	PROD39	STORE4	2	559	0.25	838.85	David Patterson	Portable Power Bank	Alexander Group
18-03-24	CUST1239	PROD60	STORE13	4	514	0.04	1973.61	Maria Hill	Luxury Perfume Spray	Gardner, Humphrey and Contreras
05-07-25	CUST1144	PROD72	STORE17	3	986	0.27	2159.43	Justin Cross	Professional Jump Rope	Holloway Ltd
04-06-24	CUST1396	PROD43	STORE18	1	590	0.22	459.86	Alexander Byrd	USB External Hard Drive	Guerrero-Schmitt
12-04-24	CUST1476	PROD69	STORE10	3	979	0.03	2850.11	Elizabeth Foster	Men's Sports Shorts	Hill, Wagner and Ballard
23-06-24	CUST1368	PROD12	STORE8	1	719	0.08	661.74	Miranda King	Hair Strengthening Shampo	Bean, Jones and Park
08-09-25	CUST1186	PROD67	STORE6	3	78	0.07	216.50	Jennifer Pruitt	Smart LED Light Strip	Benson-Elliott
17-09-24	CUST1381	PROD5	STORE3	4	853	0.14	2933.32	Erika Padilla	Men's Leather Belt	Johnson-Wright

Vlookup Using and Creating Table

Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	TOTAL AMOUNT	Customer Name	Product_Name	Store Name
07-07-24	CUST1409	PROD97	STORE3	1	218	0.09	198.42	Sarah Smith	Cricket Bat	Johnson-Wright
25-12-23	CUST1467	PROD74	STORE17	3	104	0.27	228.79	Michael Foley	Bluetooth Car Audio Syster	Holloway Ltd
03-06-24	CUST1056	PROD49	STORE8	4	914	0.25	2742.15	Christopher Sutton	Men's Roll-On Perfume	Bean, Jones and Park
29-11-23	CUST1346	PROD76	STORE8	1	922	0.16	774.07	John Jordan	Wireless Mouse	Bean, Jones and Park
06-01-24	CUST1493	PROD59	STORE5	3	745	0.18	1833.44	Amanda Powell	Men's Hair Styling Gel	Gray Ltd
26-07-24	CUST1116	PROD15	STORE7	3	147	0.19	357.23	Anthony Phillips	4K Home Projector	Martinez-Mullins
12-07-25	CUST1016	PROD56	STORE4	2	785	0.15	1335.03	John Scott	Premium Blanket Throw	Alexander Group
18-11-23	CUST1152	PROD70	STORE8	4	342	0.23	1054.56	Patrick Roberts	Kitchen Storage Rack	Bean, Jones and Park
04-05-25	CUST1187	PROD99	STORE5	1	448	0.07	416.87	Dr. Suzanne Bullock	Stainless Steel Cookware Se	Gray Ltd
31-07-25	CUST1464	PROD33	STORE0	2	873	0.23	1344.68	David Pierce	Modern Wall Clock	Rivera-Warner
13-02-24	CUST1065	PROD24	STORE14	1	279	0.07	259.13	Michael Thomas	Foldable Camping Chair	Mcdonald Inc
24-05-25	CUST1320	PROD39	STORE1	4	287	0.17	952.57	Jessica Gonzalez	Portable Power Bank	Martinez, Adkins and Johnson
24-08-24	CUST1237	PROD88	STORE4	4	645	0.01	2552.54	Wesley Short	Sports Wrist Sweatbands	Alexander Group
06-08-25	CUST1443	PROD34	STORE16	2	317	0.30	443.66	Adam Castro	Gaming Mechanical Keyboa	Frederick-Wright
21-06-25	CUST1348	PROD24	STORE15	4	403	0.29	1145.85	Sherry Cabrera	Foldable Camping Chair	Shelton PLC
10-12-24	CUST1435	PROD36	STORE9	1	871	0.04	836.49	Lindsay Beasley	Women's Summer Dress	Moore, Padilla and Butler
13-09-25	CUST1276	PROD65	STORE19	2	452	0.25	677.97	Arthur Warren	Adjustable Dumbbells 20kg	Mann-Austin
11-12-24	CUST1079	PROD27	STORE4	2	503	0.23	774.14	Molly Gardner	Decorative Floor Vase	Alexander Group
13-08-24	CUST1094	PROD39	STORE4	2	559	0.25	838.85	David Patterson	Portable Power Bank	Alexander Group
18-03-24	CUST1239	PROD60	STORE13	4	514	0.04	1973.61	Maria Hill	Luxury Perfume Spray	Gardner, Humphrey and Contreras
05-07-25	CUST1144	PROD72	STORE17	3	986	0.27	2159.43	Justin Cross	Professional Jump Rope	Holloway Ltd
04-06-24	CUST1396	PROD43	STORE18	1	590	0.22	459.86	Alexander Byrd	USB External Hard Drive	Guerrero-Schmitt
12-04-24	CUST1476	PROD69	STORE10	3	979	0.03	2850.11	Elizabeth Foster	Men's Sports Shorts	Hill, Wagner and Ballard
23-06-24	CUST1368	PROD12	STORE8	1	719	0.08	661.74	Miranda King	Hair Strengthening Shampo	Bean, Jones and Park
08-09-25	CUST1186	PROD67	STORE6	3	78	0.07	216.50	Jennifer Pruitt	Smart LED Light Strip	Benson-Elliott
17-09-24	CUST1381	PROD5	STORE3	4	853	0.14	2933.32	Erika Padilla	Men's Leather Belt	Johnson-Wright

SALES FACT

Product_ID	Store_ID	Quantity	Unit_Price	Discount	TOTAL AMOUNT
PROD97	STORE3	1	218	0.09	198.42
PROD74	STORE17	3	104	0.27	228.79
PROD49	STORE8	4	914	0.25	2742.15
PROD76	STORE8	1	922	0.16	774.07
PROD59	STORE5	3	745	0.18	1833.44
PROD15	STORE7	3	147	0.19	357.23
PROD56	STORE4	2	785	0.15	1335.03
PROD70	STORE8	4	342	0.23	1054.56
PROD99	STORE5	1	448	0.07	416.87
PROD33	STORE0	2	873	0.23	1344.68
PROD24	STORE14	1	279	0.07	259.13
PROD39	STORE1	4	287	0.17	952.57
PROD88	STORE4	4	645	0.01	2552.54
PROD34	STORE16	2	317	0.30	443.66
PROD24	STORE15	4	403	0.29	1145.85
PROD36	STORE9	1	871	0.04	836.49
PROD65	STORE19	2	452	0.25	677.97
PROD27	STORE4	2	503	0.23	774.14
PROD39	STORE4	2	559	0.25	838.85
PROD60	STORE13	4	514	0.04	1973.61
PROD72	STORE17	3	986	0.27	2159.43
PROD43	STORE18	1	590	0.22	459.86
PROD69	STORE10	3	979	0.03	2850.11
PROD12	STORE8	1	719	0.08	661.74
PROD67	STORE6	3	7	0.07	216.50

Plot Area

Find/Remove Duplicates, Handle Missing/Inconsistent Data, Correct Formatting errors, and Validate Data

REMOVE DUPLICATE DATA: Data>Data Tool>Remove Duplicates

QUANTITY, UNIT PRICE AND DISCOUNT COLUMNS HANDLE MISSING DATA with AVERAGE

Quantity: IF(ISBLANK(F2),AVERAGE(F2:F2001),F2) Unit price:IF(ISBLANK(G2),AVERAGE(G2:G2001),G2)
DISCOUNT:IF(ISBLANK(H2),AVERAGE(H2:H2001),H2)

TOTAL AMOUNT CALCULATION: Quantity*Unit Price*(1-Discount)

STORE DIM

Store_ID	Store_Name	Region	City	Store_Type
STORE0	Rivera-Warner	East	Washingtonfurt	Flagship
STORE1	Martinez, Adkins and Johnson	North	Danielmouth	Flagship
STORE2	Black-Horn	South	North Nancyview	Outlet
STORE3	Johnson-Wright	North	Hoodberg	Outlet
STORE4	Alexander Group	South	Port Sharon	Online
STORE5	Gray Ltd	North	New Michaelchester	Outlet
STORE6	Benson-Elliott	East	Port Josephberg	Online
STORE7	Martinez-Mullins	South	Williamsfurt	Online
STORE8	Bean, Jones and Park	West	Thompsonborough	Online
STORE9	Moore, Padilla and Butler	South	East Johnstad	Flagship
STORE10	Hill, Wagner and Ballard	East	East Davidville	Online
STORE11	Aguirre-Keller	South	Lake Michael	Online
STORE12	Thomas, Bridges and Harris	North	Sabrinatown	Online
STORE13	Gardner, Humphrey and Contre	East	Johnsonville	Flagship
STORE14	Mcdonald Inc	South	Christopherland	Outlet
STORE15	Shelton PLC	North	East Jenniferstad	Flagship
STORE16	Frederick-Wright	West	New Jimmymouth	Flagship
STORE17	Holloway Ltd	North	Bensonhaven	Flagship
STORE18	Guerrero-Schmitt	East	East Stevechester	Flagship
STORE19	Mann-Austin	East	Vasquezborough	Online

Plot Area

Find/Remove Duplicates, Handle Missing/Inconsistent Data, Correct Formatting errors, and Validate Data

REMOVE DUPLICATE DATA: Data>Data Tool>Remove Duplicates

CORRECT FORMATTING: PROPER(TRIM(CLEAN(B2)
PROPER(TRIM(CLEAN(D2)

PRODUCT DIM

Product_ID	Product_Name	Category	Sub_Catego	Brand	Cost	Stock
PROD0	Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	416
PROD1	Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	181
PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc	254.62	352
PROD3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	452
PROD4	Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	487
PROD5	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzal	261.00	59
PROD6	Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.80	350
PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	170
PROD8	Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	142
PROD9	Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	323
PROD10	Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	308
PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weave	449.40	271
PROD12	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	460
PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453
PROD14	Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowm	48.01	439
PROD15	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	301
PROD16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256
PROD17	Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	144.53	367
PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix	245.78	343
PROD19	Knee Support Brace	Sports	Equipment	May-Reese	21.34	65
PROD20	Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	131.51	264
PROD21	Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	123.46	76
PROD22	Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda and Rodrigu	484.62	432
PROD23	Men's Formal Shirt	Clothing	Tops	Moon, Mora and Santos	248.82	360
PROD24	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	340
PROD25	Women's Scarf Set	Clothing	Plot Area essories	Brown and Sons	174.02	268

Find/Remove Duplicates, Handle Missing/Inconsistent Data, Correct Formatting errors, and Validate Data

REMOVE DUPLICATE DATA: Data>Data Tool>Remove Duplicates

COST AND STOCK COLUMN HANDLE MISSING DATA:
COST: IF(ISBLANK(F2),AVERAGE(F2:F103),F2)
STOCK:IF(ISBLANK(G2),AVERAGE(G2:G103),G2)

CORRECT FORMATTING: PROPER(TRIM(CLEAN(B2)
PROPER(TRIM(CLEAN(C2) PROPER(TRIM(CLEAN(D2)
PROPER(TRIM(CLEAN(E2)

CUSTOMER DIM

Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	UNKNOWN
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Mar	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	UNKNOWN
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA	Silver
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA	Bronze

Find/Remove Duplicates, Handle Missing/Inconsistent Data, Correct Formatting errors, and Validate Data

REMOVE DUPLICATE DATA: Data>Data Tool>Remove Duplicates

LOYALTY_LEVEL COLUMN HANDLE MISSING DATA:
IF(ISBLANK(H2),"UNKNOWN",H2)

CORRECT FORMATTING: PROPER(TRIM(CLEAN(B2)
PROPER(TRIM(CLEAN(E2) PROPER(TRIM(CLEAN(F2)
PROPER(TRIM(CLEAN(H2)

Visualization and Insights Arrived with screenshot of visualization Report/Dashboard screenshot:

- **Descriptive analysis:**
    - Gender Wise Sales: Female customers account for 57% (1,229,710) of total sales, Male customers account for 43% (940,382).
    - Category Wise Total Sales: The Sports category recorded highest sales at 542,010, followed by 'Home' and 'Electronics' categories.
    - Regional Sales Performance (2023 - 2025): Compared to 2023, sales in the North and South regions increased significantly in 2024 and 2025. The West region recorded relatively lower sales performance.
    - Loyalty Level Quantity Sales: The majority of product volume was sold to Silver (24%), Gold (24%), and Bronze (21%) loyalty members.

- **Diagnostic Analysis**:
    - Higher Female Sales: The high sales volume in Beauty, Clothing, and Home categories primarily contributed to the higher proportion of female shoppers (57%).
    - Higher Store Wise Revenue: Online and Flagship stores generated higher revenue compared to Outlets due to a higher quantity of items purchased per order.

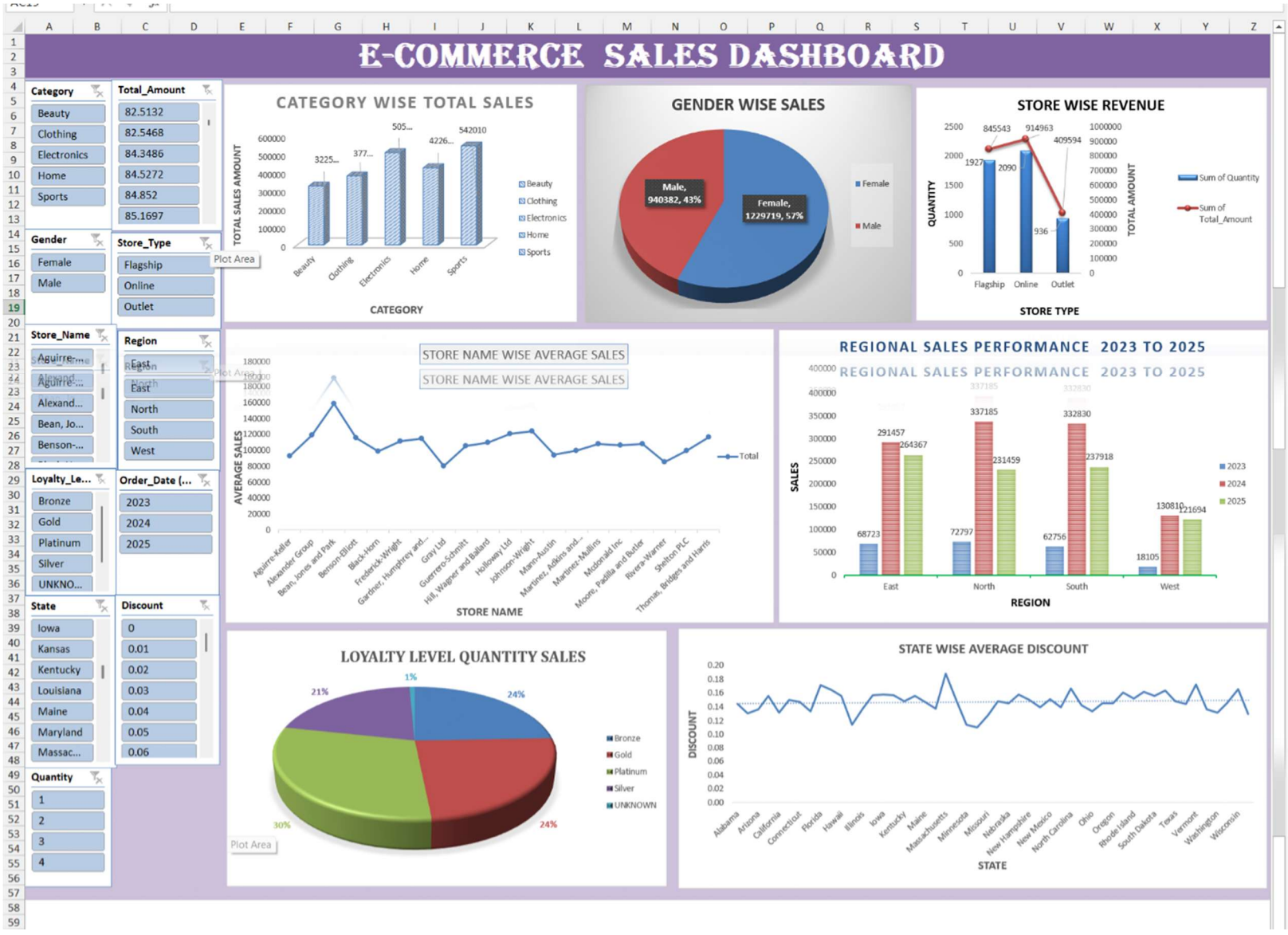
- Sales Increase (2024–2025): The growth in regional sales from 2023 to 2025 can be due to active Loyalty Programs and state specific discounts offered during this period.

- **Predictive Analysis:**
  - Sales Growth Trend: Based on historical trends from 2023 to 2025, sales in the North and South regions are projected to continue their upward growth.
  - Customer Segment Demand: The number of Silver and Gold loyalty members, along with their purchasing volume, is expected to expand further.
  - Category Demand: The Sports and Home categories are maintain strong demand in the upcoming quarters.

- **Prescriptive Analysis:**
  - West Region Focus: Since sales are lower in the West region, targeted marketing campaigns and region-specific discount offers should be introduced to boost performance.
  - Targeting Male Shoppers: To increase male sales (currently at 43%), exclusive product lines and tailored promotions for men should be introduced.
  - Inventory Management: Stock levels for high-performing categories Sports and Home must be regularly maintained to prevent stock outs.

Dashboard Image:

E-COMMERCE SALES DASHBOARD



Final 4 type of insights as storytelling:

Analyzing our e-commerce performance that our growth is powered by our Online Store platform and Female Customers (57%). Top-selling categories like Sports and Home continuously generate volume, while North and South regions lead total revenue generation. By maintaining a consistent discount strategy around 14%–16% and fostering repeat buyers through Bronze and Platinum loyalty, our business model successfully balances high customer acquisition with stable profitability.

Conclusion:

Expand Online Advertising: Allocate more budget to digital ads for online shoppers and top performing product categories.

Optimize Low Performing Regions: Introduce targeted promotions in East and West regions to match North and South regional performance.

Enhance Loyalty Rewards: Provide special offer for Bronze and Platinum members to maximize repeat order value.